COMPUTER NETWORKS

Top 50 Interview Questions & Answers

Ultimate Preparation Guide (English Edition)

This comprehensive guide is tailored to help you clear core and advanced computer networking concepts during technical interviews. It covers everything from fundamental network models to security and real-world troubleshooting scenarios.

Designed for: Software Engineers, Network Engineers, and System Administrators

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1. OSI & TCP/IP Models

Q1. Name the 7 layers of the OSI Model and state their main functions?

The OSI (Open Systems Interconnection) model consists of 7 distinct layers:

1. **Physical Layer (Layer 1):** Handles the transmission of raw bits over a physical medium (cables, fiber, radio).
2. **Data Link Layer (Layer 2):** Provides node-to-node data transfer, handles physical addressing (MAC address), and ensures error-free frame delivery. (Devices: Switch, Bridge).
3. **Network Layer (Layer 3):** Manages logical addressing (IP address) and routes packets across different networks from source to destination. (Device: Router).
4. **Transport Layer (Layer 4):** Ensures reliable end-to-end communication, session flow control, error checking, and segmentation (TCP/UDP).
5. **Session Layer (Layer 5):** Establishes, manages, maintains, and terminates sessions/connections between applications on multiple devices.
6. **Presentation Layer (Layer 6):** Translates, encrypts, decrypts, and compresses data, formatting it correctly for the application layer.
7. **Application Layer (Layer 7):** The interface where users directly interact with network applications. (Protocols: HTTP, FTP, SMTP, DNS).

Q2. What is the main difference between the OSI Model and the TCP/IP Model?

- The OSI model is a theoretical 7-layer reference model developed by the ISO, whereas the TCP/IP model is a practical 4-layer model (Network Access, Internet, Transport, Application) used in the real-world architecture of the Internet.
- OSI layers are cleanly isolated and protocol-independent, whereas TCP/IP was engineered directly around specific internet protocols.

Q3. At which layers of the OSI model do Routers, Switches, and Hubs operate?

- **Hub:** Operates at the Physical Layer (Layer 1). It blindly broadcasts incoming electronic signals to all its connected ports.
- **Switch:** Operates at the Data Link Layer (Layer 2). It routes frames to specific ports dynamically based on hardware MAC addresses.
- **Router:** Operates at the Network Layer (Layer 3). It forwards packets between independent networks using IP addresses.

Q4. Explain Data Encapsulation and Decapsulation.

- **Encapsulation:** As data moves down from the Application layer to the Physical layer on the sender's side, each layer wraps the payload with its own specialized control header (and trailer at Layer 2).
- **Decapsulation:** On the receiver's side, as raw signals move up from the physical layer to the application layer, each layer strips off its respective header to extract the underlying data.

Q5. What is the difference between a Layer 2 and a Layer 3 switch?

- **Layer 2 Switch:** Switches frames strictly within the same local network (LAN) segment by reading hardware MAC addresses. It cannot handle routing.
- **Layer 3 Switch:** Combines the speed of a switch with the intelligence of a router. It can read both MAC and IP addresses, enabling it to perform high-speed Inter-VLAN routing inside local enterprise environments.

Q6. What is a PDU (Protocol Data Unit)? Name the PDUs at different layers.

A PDU represents data with specific headers appended at a given layer of the stack:

- **Segments:** Transport Layer (Layer 4)
- **Packets:** Network Layer (Layer 3)
- **Frames:** Data Link Layer (Layer 2)
- **Bits:** Physical Layer (Layer 1)

Q7. What are the default port numbers for common Application Layer protocols like HTTP, FTP, and SMTP?

Frequently asked port standards:

- HTTP: Port 80 | HTTPS: Port 443
- FTP: Port 20 (Data Transfer), Port 21 (Control Connection) | SMTP: Port 25
- DNS: Port 53 | DHCP: Port 67 (Server), Port 68 (Client)
- SSH: Port 22 | Telnet: Port 23

2. IP Addressing & Subnetting

Q8. What are the major differences between IPv4 and IPv6?

Feature	IPv4	IPv6
Address Size	32-bit address	128-bit address
Format	Dotted Decimal (e.g., 192.168.1.1)	Hexadecimal Colon (e.g., 2001:db8::1)
Address Space	Approx. 4.3 Billion (~2^{32})	Practically infinite (~2^{128})
Security (IPSec)	Optional / Add-on	Built-in / Mandated

Q9. Differentiate between a MAC Address and an IP Address.

- **MAC Address:** A 48-bit permanent physical/hardware address burned onto a device's Network Interface Card (NIC) during manufacturing. Used for local node delivery.
- **IP Address:** A logical, temporary address assigned by network administrators or via DHCP (32-bit for IPv4, 128-bit for IPv6). Used for routing packets globally across networks.

Q10. What is Subnetting and why is it required?

Subnetting is the architectural practice of breaking down a large, single physical IP network into multiple smaller, logically isolated subnetworks.
Requirement: It reduces IP address wastage, minimizes network congestion by shrinking broadcast domains, and significantly hardens network security by isolating subnets.

Q11. What is the difference between Classful and Classless (CIDR) IP addressing?

- **Classful Addressing:** Segments the IP pool into rigid Classes (A, B, C, D, E) with fixed, immutable network masks (/8, /16, /24). It leads to severe IP wastage.
- **Classless Addressing (CIDR):** Completely eliminates rigid class boundaries. It uses arbitrary mask lengths specified via slash notation (e.g., 10.0.0.0/22), matching resources perfectly to requirements.

Q12. What is the difference between Private IP and Public IP addresses?

- **Private IP:** Allocated for internal corporate/home LAN usage and is completely unroutable over the public internet. They are free to use.
- **Public IP:** Globally unique addresses assigned by registries/ISPs that are explicitly routable over the public internet. They are paid resources.

Q13. What is a Loopback Address and what is its primary use?

A loopback address (IPv4: **127.0.0.1**, IPv6: **::1**) maps back to the local host machine. It is used to test the local system's internal TCP/IP network stack. Running `ping 127.0.0.1` ensures that the network stack is initialized and running properly on your computer.

Q14. What does a Subnet Mask represent? How do you derive Network ID and Host ID?

A subnet mask is a 32-bit filter that identifies which bits of an IP address belong to the Network ID (indicated by binary 1s) and which belong to the Host ID (indicated by binary 0s).
To find the Network ID, a bitwise logical AND operation is computed between the target IP and its subnet mask. The remaining unmasked portion represents the individual Host ID.

Q15. What is NAT (Network Address Translation) and why is it used?

NAT is a technique implemented on boundary gateways/routers that maps multiple unroutable Private IP addresses within a local LAN to a single public IP address before forwarding packets out to the public internet. This conserves the scarce public IPv4 address space.

3. Transport Layer & Protocols (TCP vs UDP)

Q16. What is the difference between TCP and UDP? Give real-world use cases.

- **TCP (Transmission Control Protocol):** A connection-oriented protocol that guarantees delivery via sequence tracking, strict error checking, and packet acknowledgements. Use Cases: Web Browsing (HTTP), File Transfers (FTP), Email (SMTP).
- **UDP (User Datagram Protocol):** A lightweight, connectionless protocol that sends datagrams instantly without waiting for a connection or verifying receipts. Use Cases: Video Streaming, Online Gaming, Voice calls (VoIP), DNS queries.

Q17. Explain the TCP Three-Way Handshake mechanism.

Before transmitting any data, TCP establishes a verified channel using 3 specific steps:

1. **SYN (Synchronize):** The client transmits a packet containing a random Initial Sequence Number (ISN) with the SYN flag set to request a connection.
2. **SYN-ACK:** The server responds with its own sequence number, setting both the SYN and ACK flags to confirm receipt of the client's packet.
3. **ACK:** The client transmits a final ACK packet back to confirm. The connection status changes to "Established."

Q18. How do Flow Control and Congestion Control work in TCP?

- **Flow Control:** Prevents a fast sender from overwhelming a slow receiver's memory buffers. It uses a dynamic **Sliding Window** buffer advertisement technique.
- **Congestion Control:** Prevents the broader network infrastructure from collapsing due to excessive traffic. It dynamically alters speed using algorithms like **Slow Start, Congestion Avoidance, Fast Retransmit**.

Q19. What is Windowing (Sliding Window Protocol)?

Windowing defines the specific quantity of data bytes a sender can transmit over the line before it is forced to stop and wait for an acknowledgement packet from the receiver. This allows optimal pipeline utilization of link bandwidth.

Q20. Why do we prefer UDP over TCP for real-time video calls despite packet loss?

In live video streams, low latency and speed are critical. If a packet drops under TCP, it pauses the live stream to retransmit the missing data, causing severe lag. UDP continuously streams new frames without pausing. A minor dropped packet might cause a temporary glitch, but the live connection remains smooth.

Q21. Explain how a TCP connection is terminated (Four-Way Handshake).

TCP closes active connections gracefully using a 4-step sequence:

1. The active device sends a **FIN** flag packet indicating it has no more data to send.
2. The receiving side responds with an **ACK** packet and finishes its pending data transmission.
3. Once ready to close, the receiving side transmits its own **FIN** packet.
4. The initiating client returns a final **ACK**, entering a brief timeout wait before fully releasing resources.

4. Application Layer Protocols & Web

Q22. What is DNS (Domain Name System) and how does it function?

DNS acts as the "phonebook of the Internet." It translates human-readable web domains (e.g., google.com) into numeric machine IP addresses (e.g., 142.250.190.46). It resolves requests hierarchically by polling Root Nameservers, Top-Level Domain (TLD) servers, and Authoritative Nameservers.

Q23. What is the difference between Recursive and Iterative DNS queries?

- **Recursive Query:** The client forces the local DNS resolver to handle the entire search. The resolver queries external servers until it returns either a valid IP or a clear error message.
- **Iterative Query:** A DNS server returns the best answer it currently knows or provides a referral to another nameserver down the chain, leaving the resolver to manage the next step.

Q24. Differentiate between HTTP and HTTPS. What is an SSL/TLS handshake?

- **HTTP:** Transmits web data in plain text. It is vulnerable to eavesdropping (Unsecure, Port 80).
- **HTTPS:** Secures data transmissions by wrapping HTTP in an encrypted layer (Secure, Port 443).
- **SSL/TLS Handshake:** A session initialization routine where a client validates a server's identity certificate and negotiates cryptographic symmetric keys to encrypt session traffic.

Q25. How does DHCP work? Explain the DORA process.

DHCP automates the assignment of IP configurations to joining network hosts via the 4-step **DORA** cycle:

- **Discover (D):** The unconfigured client broadcasts a packet to locate any active DHCP servers.
- **Offer (O):** Active servers respond with an unallocated IP address proposal.
- **Request (R):** The client broadcasts a request to reserve the offered configuration parameters.
- **Acknowledgment (A):** The server confirms the reservation and grants a lease lease to the client.

Q26. What is the difference between FTP and TFTP?

- **FTP:** Robust, connection-oriented (TCP-based, ports 20/21) protocol supporting user logins, encryption options, and advanced command controls. Optimized for moving large application payloads.
- **TFTP:** A barebones, lightweight, connectionless protocol (UDP-based, port 69). It lacks authentication and is typically used to boot diskless workstations or push network firmware images.

Q27. Contrast SMTP, IMAP, and POP3 email protocols.

- **SMTP:** Used exclusively to push/route outgoing emails between local clients and external destination mail servers.
- **POP3:** Downloads mail directly onto a single offline device and typically deletes it from the server, making multi-device sync difficult.
- **IMAP:** Keeps messages stored centrally on the mail server while syncing state changes across multiple concurrent user devices.

Q28. What is the difference between ARP and RARP?

- **ARP (Address Resolution Protocol):** Resolves a known destination IP address into its physical MAC address inside local subnets.
- **RARP (Reverse ARP):** Resolves a known hardware MAC address into an IP address assignment (now replaced by DHCP capabilities).

5. Routing & Switching Concepts

Q29. What is the fundamental difference between Routing and Switching?

- **Switching:** Connects individual systems within the *same* local area network segment (LAN) and moves traffic based on Layer 2 MAC addresses.
- **Routing:** Interconnects *completely different* networks and steers packets globally across boundaries using Layer 3 IP destinations.

Q30. Explain Unicast, Multicast, and Broadcast traffic patterns.

- **Unicast:** Standard point-to-point communication (One-to-One).
- **Multicast:** Direct stream to a specific, subscribed sub-group of listening stations (One-to-Many).
- **Broadcast:** Transmits a single stream to every listening host on the local domain (One-to-All).

Q31. What is a Broadcast Storm and how is it mitigated?

A broadcast storm occurs when redundant paths exist between Layer 2 switches without loop prevention. Broadcast packets loop indefinitely, multiplying rapidly and consuming all available network bandwidth, which can cause the network to crash. This is mitigated by enabling **STP (Spanning Tree Protocol)** on the switches.

Q32. What is a VLAN (Virtual LAN) and why is it used?

A VLAN logically segments a single physical switch into separate, isolated virtual broadcast domains. It enhances security, organizes internal departments (e.g., HR, Finance) independently, and reduces broadcast traffic across the network.

Q33. What is Spanning Tree Protocol (STP) and how does it prevent switching loops?

STP (IEEE 802.1D) prevents network loops by mapping all Layer 2 paths and dynamically blocking redundant links. If a primary path fails, STP automatically opens the blocked backup port to restore connectivity without creating loops.

Q34. What is a Routing Table and what information does it contain?

A routing table is a database stored in a router's RAM that lists optimal paths to destination networks. Key entries include:

- Target Network Address and Subnet Mask
- Next-Hop IP address (the next gateway router)
- Routing metric/cost value (to evaluate path quality)
- Outbound Physical Interface

Q35. Differentiate between Static Routing and Dynamic Routing.

- **Static Routing:** Paths are manually configured by a network administrator. It is highly secure and has zero CPU overhead, making it ideal for small, stable networks.
- **Dynamic Routing:** Routers use discovery protocols (e.g., OSPF, BGP) to learn paths automatically, exchange updates, and dynamically calculate alternate routes if a link fails. Optimized for large, complex networks.

Q36. What is the difference between IGP and EGP protocols?

- **IGP (Interior Gateway Protocol):** Used to route packets within a single company or Autonomous System. Examples include OSPF, EIGRP, and RIP.
- **EGP (Exterior Gateway Protocol):** Used to exchange routing information between entirely separate Autonomous Systems. **BGP** is the primary EGP that routes traffic across the global Internet.

Q37. Compare Distance Vector and Link State routing methodologies.

- **Distance Vector (e.g., RIP):** Routes traffic based strictly on hop count. It periodically broadcasts its entire routing table to directly connected neighbors, which can cause slow convergence.
- **Link State (e.g., OSPF):** Builds a complete topology map of the entire network area using Dijkstra's Shortest Path First algorithm. It only broadcasts incremental updates when changes occur, leading to fast convergence and efficient bandwidth use.

Q38. What is BGP (Border Gateway Protocol)?

BGP is a Path-Vector routing protocol that acts as the core routing engine of the global Internet. It manages routing decisions between independent networks and Autonomous Systems (ISPs, universities, large tech companies) using policy rules rather than raw bandwidth speed metrics.

6. Network Security & Advanced Concepts

Q39. What is a Firewall? Differentiate between Packet Filtering and Stateful Inspection.

A firewall is a network security device that monitors and filters traffic based on predefined security rules.

- **Packet Filtering (Stateless):** Inspects individual packets independently based on basic criteria like Source/Destination IP and Port numbers, without tracking the overall connection state.
- **Stateful Inspection:** Tracks the state of active network connections. It recognizes context and only permits external traffic if it is a valid response to an internal request.

Q40. What is a VPN (Virtual Private Network) and how does it secure data traffic?

A VPN creates an encrypted, isolated tunnel across an unsecure public network (like the Internet). It uses secure encapsulation and protocols like **IPSec** or **SSL/TLS** to encrypt data packets, ensuring confidentiality and integrity even over public connections.

Q41. What is the difference between Symmetric and Asymmetric encryption?

- **Symmetric Encryption:** Uses a single shared key for both encryption and decryption. It is fast and efficient for bulk data transfer (e.g., AES).
- **Asymmetric Encryption:** Uses a mathematically linked key pair: a Public Key for encryption and a Private Key for decryption. It provides high security for key exchange but is computationally intensive (e.g., RSA).

Q42. What is a Proxy Server and how does it differ from a Reverse Proxy?

- **Forward Proxy:** Sits in front of internal clients to filter outgoing requests, hide client identities, and control internet access.
- **Reverse Proxy:** Sits in front of backend web servers to route incoming client requests. It handles tasks like load balancing, caching, and SSL decryption to protect the backend infrastructure.

Q43. Explain DoS and DDoS attacks.

- **DoS (Denial of Service):** An attack from a single system that floods a target server with fake traffic to overwhelm its resources and crash its services for legitimate users.
- **DDoS (Distributed DoS):** A coordinated attack from thousands of distributed, compromised devices (botnets) targeting a server simultaneously, making it difficult to isolate and block the attack traffic.

Q44. Differentiate between Spoofing and Phishing.

- **Spoofing:** A technique where an attacker fakes data (like modifying an IP header or MAC address) to impersonate a trusted system and bypass security controls.
- **Phishing:** A social engineering attack that uses deceptive emails or spoofed websites to trick users into revealing sensitive data, such as passwords or credit card details.

Q45. What is the difference between an IDS and an IPS?

- **IDS (Intrusion Detection System):** A passive security system that monitors network traffic and alerts administrators when it detects suspicious activity, without taking corrective action.
- **IPS (Intrusion Prevention System):** An active security system that monitors traffic and automatically blocks or drops malicious data streams as soon as a threat is identified.

7. Troubleshooting & Practical Scenarios

Q46. Which protocol is used by 'ping' and 'traceroute'? What is the difference between these commands?

Both commands rely on **ICMP (Internet Control Message Protocol)**.

- **Ping:** Uses ICMP Echo Request and Echo Reply messages to verify if a destination host is reachable and measures round-trip time latency.
- **Traceroute:** Maps the exact path packets take to a destination. It increments the **TTL (Time to Live)** field in packet headers from 1 upward, causing each intermediate router to return an ICMP Time Exceeded message, revealing every hop along the route.

Q47. If you can successfully ping a website's IP address but cannot open it in a web browser, what could be the issue?

- The target web server application may be down or rejecting connections on web ports (Port 80/443), even though its low-level network stack answers ICMP pings.
- A firewall along the path or on your system may be blocking outbound HTTP/HTTPS traffic.
- Your browser's local cache or proxy configurations could be misconfigured.

Q48. What is a Default Gateway? What happens if it is misconfigured?

A default gateway is the local router interface IP that allows devices inside a LAN to send traffic outside their own network segment.
Misconfiguration: If configured incorrectly, your local device can still communicate with other systems on the same local network but will be unable to access external networks or the Internet.

Q49. What are the uses of the 'ipconfig' and 'ifconfig' / 'ip a' commands?

These diagnostic utilities display current network adapter configurations. They provide details such as your system's assigned IP address, Subnet Mask, Default Gateway, and physical hardware MAC address for troubleshooting network connections (use `ipconfig` on Windows, and `ifconfig` or `ip a` on Linux systems).

Q50. Step-by-step, what happens behind the scenes when you type 'https://www.google.com' into a browser and press Enter?

1. **URL Parsing:** The browser extracts the host name and identifies the protocol as secure HTTPS.
2. **DNS Resolution:** The browser checks its local cache, host files, and queries recursive DNS resolvers to map the name `www.google.com` to its corresponding public IP address.
3. **TCP Handshake:** The client initiates a TCP Three-Way Connection with the destination server on Port 443 (SYN -> SYN-ACK -> ACK).
4. **TLS/SSL Handshake:** The client validates the server's security certificate and establishes an encrypted symmetric session key.
5. **HTTP GET Request:** The browser transmits an encrypted HTTP request to fetch the website's index page data.
6. **Server Response & Rendering:** Google's servers process the request and return the web payload. The browser decrypts, parses, and renders the HTML, CSS, and Javascript code on your screen.